SOUMENDRA GANGULY

Aarhus, Denmark
+45 50 36 19 83
soumendraganguly@gmail.com
github.com/8vasu
soumendraganguly.com
linkedin.com/in/soumendraganguly
arXiv.org

SKILLS

Main Languages: **C, Python**

*NIX sysadmin/netadmin: **Linux, Bash, FreeBSD, POSIX shell**

DevOps: **Git, GitHub Actions (CI/CD), Docker**

Productivity: **GNU Emacs, Emacs Lisp, Org Mode**

Math: **SageMath, SymPy, NumPy, CVXPY, Matplotlib, Lark parser**

AI and Machine Learning: **OpenAI API, PyTorch, Stable Diffusion**

Data: **PostgreSQL, Psycopg, pandas, SQLAlchemy, TimescaleDB, Redis**

Front-end: **HTML, CSS, Bootstrap, Hugo, Jinja, JavaScript (ECMAScript)**

Back-end: **Django, Flask, Apache, NGINX, Gunicorn**

Typesetting: LaTeX, **Lua, PGF/TikZ, AUCTeX**

PROJECTS

- github.com/8vasu/sg-diffusion
  NVIDIA NGC PyTorch + Stable Diffusion.
- github.com/8vasu/power-opsd-dk1
  Allocate from Danish electricity prices/hour.
- github.com/8vasu/paper.py
  Command-line interface for arXiv.org.
- github.com/8vasu/compuTeX
  GNU Emacs – LaTeX ↔ SymPy.
- github.com/8vasu/2windows.el
  GNU Emacs – convenient 2-window system.
- github.com/8vasu/8vasu.github.io
  Source for soumendraganguly.com.
- github.com/8vasu/stty.py
  pypi.org/project/stty
  POSIX *stty(1)*-style Python library.
- github.com/8vasu/fretplot
  ctan.org/pkg/fretplot
  Easy guitar tabs in LaTeX (coming soon).

EDUCATION

2019–2023, Texas A&M University
PhD, Mathematics

2017–2019, Clemson University
MS, Mathematical Sciences

2014–2016, Chennai Mathematical Ins.
MSc, Mathematics

2011–2014, Chennai Mathematical Ins.
BSc, Math and Computer Science

LANGUAGES

English, Hindi, Bangla ■■■■■
Danish (DU3, Module 3) ■■■

PROFILE

I am an experienced computer scientist and programmer with over fourteen years of expertise in software development, system administration, and web technologies. I am highly proficient in Linux and other *nix platforms. I have contributed significant features, implemented unit tests, and fixed bugs in several major open-source projects.

In addition to my technical background, I am a mathematician with a strong research focus on the analytic behavior of automorphic L-functions and number theory.

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher in Mathematics at Aarhus University — Sep 2023–NOW
Research

- **Problem solving, analytical reasoning, creativity, initiative:** Employed rigorous proof techniques to analyze and solve complex, abstract problems in the theory of automorphic L-functions, contributing to the understanding of their underlying structures.
- **Leadership, project management:** Principal organizer and driving force behind a 2-week international event: The Aarhus Automorphic Forms Conference and Summer School, August 4–15, Aarhus University. This involves managing a team of 9 organizers, coordinating 21 invited speakers, 12 contributed talk speakers, overseeing the logistics of the event including accommodation, transportation, dining, visa applications, and managing budget, time, and funding applications for 100+ participants from around the world.
- **Data analysis and communication:** Interpretation and presentation of data and research results to small and large groups. My academic CV is available on my website linked in the header.
- **Research:** Performed literature review and analysis of existing research papers.

Independent programmer — Dec 2023–NOW
Collaboration with SymPy

- **Feature development:** Added LaTeX matrix and complex number parsing capability to the official **SymPy** project in a significant pull request.
- **Package development:** Developed compuTeX, a GNU Emacs package enabling users to input complex mathematical expressions in LaTeX format and output the evaluated expression back in LaTeX with a single keystroke, powered by SymPy.
- **Tools:** Python, SymPy, the Lark parser with EBNF grammar, regular expressions, LaTeX , GNU Emacs, Emacs Lisp, Git.

Independent programmer — Sep 2020–NOW
Collaboration with CPython, FreeBSD, NetBSD, and util-linux

- **Feature development, long-term sustained contribution:**
  - Added 7 new functions to the *os*, *termios*, *tty*, and *pty* modules of the official **CPython** project in an ongoing GitHub issue containing 20 pull requests over 5+ years, significantly enhancing the Python standard library's functionality.
  - Added 2 functions to the standard C library of the **FreeBSD** operating system and wrote the manual page for them.
  - Improved the standard *script(1)* and *scriptreplay(1)* programs on FreeBSD, NetBSD, and util-linux as demonstration of these functionalities.
- **Bug fixing:** Addressed and resolved pseudoterminal-related issues within the CPython project.
- **Software testing:** Created comprehensive **unit tests** to validate the reliability and correctness of newly introduced features.
- **Package development:** Developed stty.py for clean, Pythonic POSIX *stty(1)*-style terminal manipulation. Published on PyPI via automated GitHub Actions workflows.
- **Tools:** C, Python, POSIX/BSD/GNU termios and pseudoterminal APIs, unit testing, Git, Docker, GNU Autotools, *nroff(1)*, *script(1)*, *scriptreplay(1)*, *stty(1)*, GitHub Actions.